Dense and lexical retrieval

Lexical retrieval compares words appearing in a question and a passage. TF-IDF increases the weight of terms frequent in a passage but uncommon across the corpus. A lexical baseline is inexpensive and interpretable, but it can miss passages that express the same idea with different vocabulary.

Dense retrieval maps text into a vector representation using a pretrained encoder. Normalize both question and passage vectors to unit length. Their dot product then equals cosine similarity. A higher cosine score indicates greater alignment in the representation space; it is not a calibrated probability that a passage answers the question.

Evaluate the lexical baseline and the dense system on the same relevance labels. Specialized terminology, identifiers, and exact names may favor lexical retrieval, while paraphrased questions can benefit from semantic representations. A dense model pretrained on general text may underperform on unfamiliar domains.

Chunking and evidence provenance

Chunking divides extracted text into windows that fit a retrieval encoder and generation context. Overlap repeats boundary words between adjacent windows, helping preserve local continuity. Larger windows include more context but can dilute a short relevant statement. Smaller windows improve granularity but can separate an answer from its explanation.

Keep windows within a page so every passage retains an unambiguous page reference. A source record includes document content hash, page number, word offsets, and chunk ID. Stable chunk IDs should depend on content and chunking parameters. Changing the chunk size can change chunk IDs even when the original PDF is unchanged.

Word counts are not tokenizer counts. Technical words may split into multiple subwords, and an embedding model may truncate a long window. Measure retrieval across several window sizes rather than assuming that one setting is universally optimal. Page labels remain usable across chunking experiments when the PDF bytes remain unchanged.

Grounded generation and abstention

Retrieval-augmented generation first retrieves passages and then asks a language model to answer from those passages. Give each retrieved passage a source label. Ask the model to cite those labels for factual claims and to state when evidence is insufficient. Keep source passages visible so readers can inspect support.

Document text is untrusted data. A passage can contain instructions that attempt to redirect the model. Place behavioral instructions outside the evidence and state that evidence instructions must not be followed. This separation reduces some risks but does not eliminate prompt injection.

Citation validation checks that referenced IDs were supplied to the model. It does not establish that a cited passage entails a claim. An answer can cite a valid source and still be wrong. Withhold answers with unknown citation labels, and evaluate claim support separately with human review. If local inference is unavailable, show retrieved evidence without inventing an answer.